Lecture 8 – More on Classification Models

[ISLP 4.4, 4.5]

MC886 – Machine Learning

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Today's summary

Linear Discriminant Analysis (LDA)

Quadratic Discriminant Analysis (QDA)

Naive Bayes

A comparison among classification methods

Generative models for classification

In logistic regression, we use the logistic function to model:

$$\Pr(Y = k | X = x)$$

Instead, we could model the distribution of X for each class:

$$\Pr(X = x | Y = k),$$

and, using the distribution of Y, to use the *Bayes' theorem* to convert them into estimates of $\Pr(Y = k | X = x)$

Bayes' theorem

Suppose that we want to classify an observation x into one of $K \geq 2$ classes. Let's also consider that:

- π_k: the prior probability that x come from the kth class

- $f_k(X) \equiv \Pr(X|Y = k)$: the density function of X for a x that comes from the kth class (likelihood)
  (for a quantitative X, consider $f_k(x)dx$ the probability of X falling in a small region dx around x)

- The Bayes' theorem states that:

$$\underbrace{\Pr(Y = k|X = x)}_{\text{the posteriori } p_k(x)} = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)} \qquad (4.15)$$

How to estimate π_k and $f_k(x)$?

- For a random sample from the population, π_k can be estimated by computing the fraction of $y_i = k$ for $i = 1, \ldots, n$ observations

- However, estimating $f_k(x)$ is trickier, and requires some simplifying assumptions

- In this lecture, we'll see three classifiers that use different simpliflying assumptions for estimating $f_k(x)$

- The first one is the **Linear Discriminant Analysis (LDA)**

LDA for $p = 1$

- Let's assume that $f_k(x)$ is *Gaussian*, whose density is:

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma_k}} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right), \qquad (4.16)$$

  where μ_k and σ_k^2 are the mean and variance for the kth class

- Let's also assume that $\sigma = \sigma_1 = \ldots \sigma_K$. This way, plugging Eq. 4.16 into Eq. 4.15 yields:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{l=1}^{K} \pi_l \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_l)^2\right)} \qquad (4.17)$$

LDA for $p = 1$

- If $f_k(x)$ is indeed Gaussian, we have a Bayes classifier, so we assign to x a class k with largest $p_k(x) = \Pr(Y = k|X = x)$

- However, we can simplify the expression: taking the log of Eq. 4.17 and rearranging terms yields:

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k), \tag{4.18}$$

  whose largest $\delta_k(x)$ is equivalent to the largest $p_k(x)$

- For $K = 2$, the Bayes decision boundary is an x where $\delta_1(x) = \delta_2(x)$:

$$x = \frac{\mu_1^2 - \mu_2^2}{2(\mu_1 - \mu_2)} = \frac{\mu_1 + \mu_2}{2} \tag{4.19}$$

Estimates of μ_k, σ^2, π_k and $\delta_k(x)$

- LDA approximates the Bayes classifier by assuming $f_k(x)$ Gaussian and estimating parameters from data:

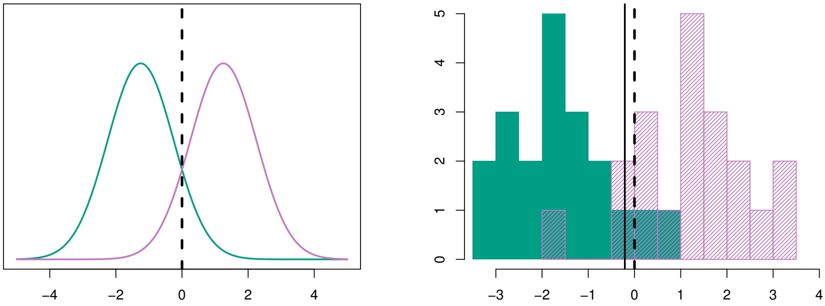
$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i$$

$$\hat{\sigma^2} = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=K} (x_i - \hat{\mu}_k)^2 \tag{4.20}$$

$$\hat{\pi}_k = \frac{n_k}{n} \tag{4.21}$$

$$\hat{\delta}_k(x) = x\frac{\hat{\mu}_k}{\hat{\sigma}^2} - \frac{\hat{\mu}_k^2}{2\hat{\sigma}^2} + \log(\hat{\pi}_k), \tag{4.22}$$

Two Gaussian density functions (left) and LDA on 40 points drawn from those functions (right)



Dashed line: Bayes decision boundary (given by $\delta_1 = \delta_2$)
Solid line: LDA decision boundary (given by $\hat{\delta}_1 = \hat{\delta}_2$)

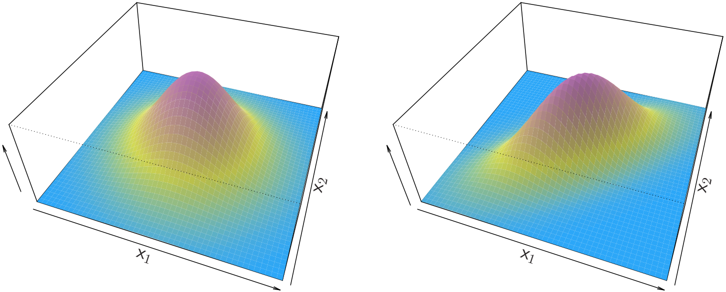
LDA for $p > 1$

- We assume $X = (X_1, X_2, \ldots, X_p)$ as a *multivariate Gaussian* distribution, with:
  - class-specific p-dimensional mean vectors μ_k
  - common $p \times p$ covariance matrix $\mathbf{\Sigma}$

- The pdf of multivariate Gaussian distribution is:

$$f(x) = \frac{1}{(2\pi)^{(p/2)}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \mathbf{\Sigma}^{-1}(x-\mu)\right) \tag{4.23}$$

Uncorrelated (left) and correlated (right) multivariate Gaussian distributions



LDA for $p > 1$

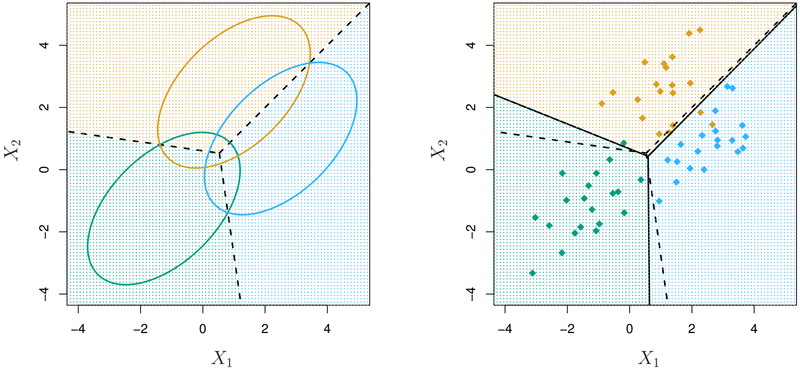
- Akin to $p = 1$, we assume that a x from the kth class is drawn from a multivariate Gaussian distribution $N(\mu_k, \mathbf{\Sigma})$

- We can plug the multivariate density function $f_k(x)$ into Eq. 4.15 and with some algebra obtain:

$$\delta_k(x) = x^T \mathbf{\Sigma}^{-1} \mu_k - \frac{1}{2} \mu_k^T \mathbf{\Sigma}^{-1} \mu_k + \log(\pi_k), \qquad (4.24)$$

  whose largest value is equivalent to the largest one of the Bayes classifier (assuming multivariate Gaussian a correct approximation for $f_i(x)$, $i = 1, \ldots, K$)

LDA for $p > 1$ and three classes



Dashed line: Bayes decision boundary (given by $\delta_k = \delta_l$)
Solid line: LDA decision boundary (given by $\hat{\delta}_k = \hat{\delta}_l$)

Confusion matrix for LDA on the Default data

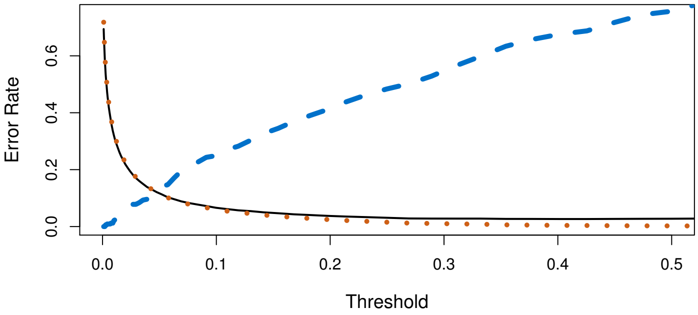
		True default status		
		No	Yes	Total
Predicted	No	9644	252	9896
default status	Yes	23	81	104
	Total	9667	333	10000

- Here we classified as default for $\Pr(\texttt{default}|X = x) > 0.5$
- How could we improve the predictions?

Confusion matrix for LDA on the Default data $(\Pr(\texttt{default}|X = x) > 0.2)$

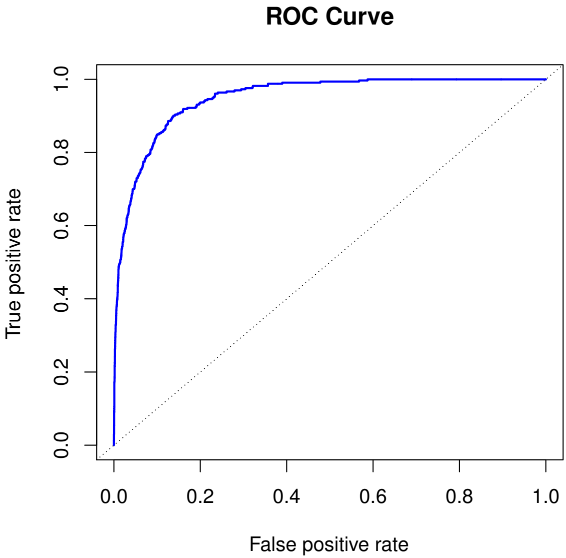
		True default status		
		No	Yes	Total
Predicted	No	9432	138	9570
default status	Yes	235	195	430
	Total	9667	333	10000

Error rate as a function of posterior probability threshold for LDA on the Default data



Dashed line: defaulting customers incorrectly classified
Dotted line: non-defaulting customers incorrectly classified
Solid line: overall error rate

ROC curve for the LDA on the Default data



Quadratic Discriminant Analysis (QDA)

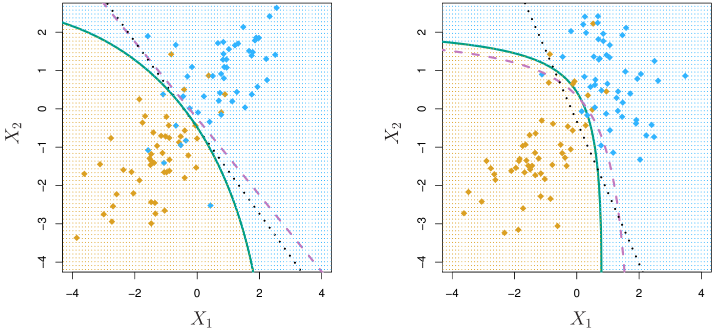
- Now we assume that an x from the kth class come is of the form $X \sim N(\mu_k, \mathbf{\Sigma}_k)$ (i.e., a correlation matrix $\mathbf{\Sigma}_k$ for each class k)

- Under this assumption, the Bayes classifier assigns an observation $X = x$ to the class for which:

$$\delta_k(x) = -\frac{1}{2}(x - \mu_k)^T \mathbf{\Sigma}_k^{-1}(x - \mu_k) - \frac{1}{2}\log|\mathbf{\Sigma}_k| + \log \pi_k, \tag{4.28}$$

is largest

QDA on a two-class problem with $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$ (left) and with $\boldsymbol{\Sigma}_1 \neq \boldsymbol{\Sigma}_2$ (right)



Dashed line: Bayes decision boundary
Dotted line: LDA decision boundary
Solid line: QDA decision boundary

Naive Bayes

- Here, we make a single assumption:

  "Within the kth class, the p predictors are independent"

- Mathematically, it is translated as:

$$f_k(x) = f_{k1}(x_1) \times f_{k2}(x_2) \times \ldots \times f_{kp}(x_p), \tag{4.29}$$

  for $k = 1, \ldots, K$

- Plugging Eq. 4.29 into Eq. 4.15 yields:

$$\Pr(Y = k|X = x) = \frac{\pi_k f_{k1}(x_1) \times f_{k2}(x_2) \times \ldots \times f_{kp}(x_p)}{\sum_{l=1}^{K} \pi_l f_{l1}(x_1) \times f_{l2}(x_2) \times \ldots \times f_{lp}(x_p)}, \tag{4.30}$$

  for $k = 1, \ldots, K$

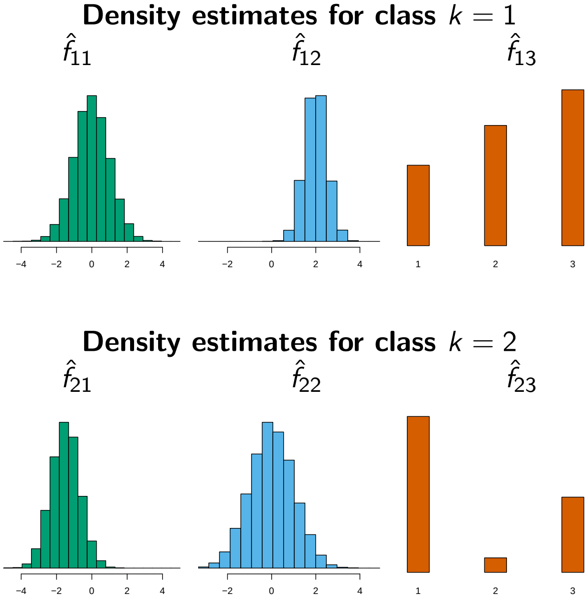
How to estimate f_{kj}?

1. X_j quantitative: $X_j | Y = k \sim N(\mu_{jk}, \sigma^2_{jk})$

2. X_j quantitative: non-parametric estimate (e.g., histogram)

3. X_j qualitative: frequency of each class

Example for $K = 2$



Confusion matrix for naive Bayes on the `Default` data ($\Pr(\texttt{default}|X = x) > 0.5$)

		True default status		
		No	Yes	Total
Predicted	No	9621	244	9865
default status	Yes	46	89	135
	Total	9667	333	10000

Confusion matrix for naive Bayes on the Default data ($\Pr(\texttt{default}|X = x) > 0.2$)

		True default status		
		No	Yes	Total
Predicted	No	9339	130	9469
default status	Yes	328	203	531
	Total	9667	333	10000

A comparison among classification methods

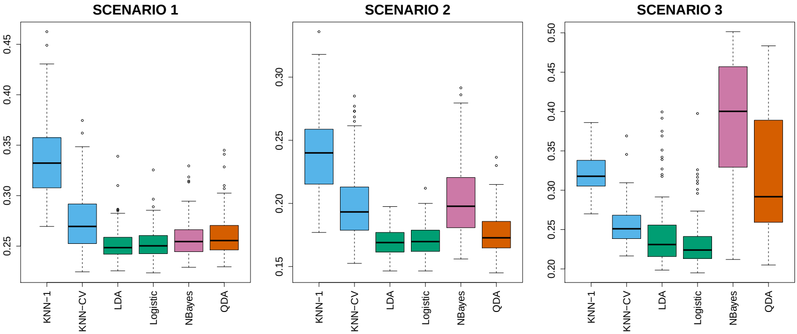
- Empirical comparison between KNN-1, KNN-CV, LDA, logistic regression, naive Bayes and QDA

- Six scenarios were considered
(let's inspect 3 of them; the remaining ones are in ISLP 4.5):

  1. $n_k = 20$ in each of two classes. Points within each class were uncorrelated random normal variables with a different mean in each class

  2. Same as 1, except that within each class, the two predictors had a correlation of -0.5

  3. Same as 2, except that it is used a t-distribution (which has fat tails), and 50 points per class


SCENARIO 1
SCENARIO 2
SCENARIO 3
KNN−1
KNN−CV
LDA
Logistic
NBayes
QDA